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0 2 Chromatography is used to separate mixtures.

Chromatography involves a mobile phase and one other phase.

0 2 . 1 What is the other phase in chromatography?

[1 mark]

Tick (✓) **one** box.

Moving phase ☐

Recycled phase ☐

Stationary phase ☐

Viscous phase ☐

0 2 . 2 Why do the substances in the mixture separate in the mobile phase?

[1 mark]

_____

_____

0 2 . 3 How many spots will be produced on the chromatogram of a pure compound?

[1 mark]

Number of spots = _____

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0 2 . 4 In a chromatography experiment, a blue colour moved 4.77 cm.

The solvent moved 5.30 cm.

Calculate the R_f value for the blue colour.

[2 marks]

_____

_____

_____

R_f value = _____

Question 2 continues on the next page

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0 2 . 5 Black ink is a mixture of several colours.

Plan an experiment using paper chromatography to:

- separate the colours in black ink
- identify the colours from their R_f values.

[6 marks]

11

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Turn over for the next question

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